Multiplication of Two Numbers Without Using the * Operator

C Language – Algorithm and Flowchart Debugging

1. Mistakes in the Given Algorithm

The given algorithm uses the statement "Repeat ... until (index <= num2)". This does not match the way the loop is shown in the flowchart. The steps need to continue while the condition is true.

Another problem is that num2 is always used as the number of repetitions. This can cause extra additions when num2 is larger than num1.

2. Corrected Algorithm

1. Define num1, num2, prod, index, add and count as integers.

2. Get num1 and num2.

3. If num1 < num2, set add = num2 and count = num1.

4. Otherwise, set add = num1 and count = num2.

5. Set prod = 0 and index = 1.

6. Repeat while index <= count:

    prod = prod + add

    index = index + 1

9. Print prod.

10. Stop.

3. Minimum Number of Additions

For positive integer inputs, the minimum number of additions is obtained by adding the larger number the smaller number of times.

Minimum additions = min(num1, num2).

For example, for 3 and 8, add 8 three times: 8 + 8 + 8 = 24. So only 3 additions are required.

4. Mistakes in the Given Flowchart

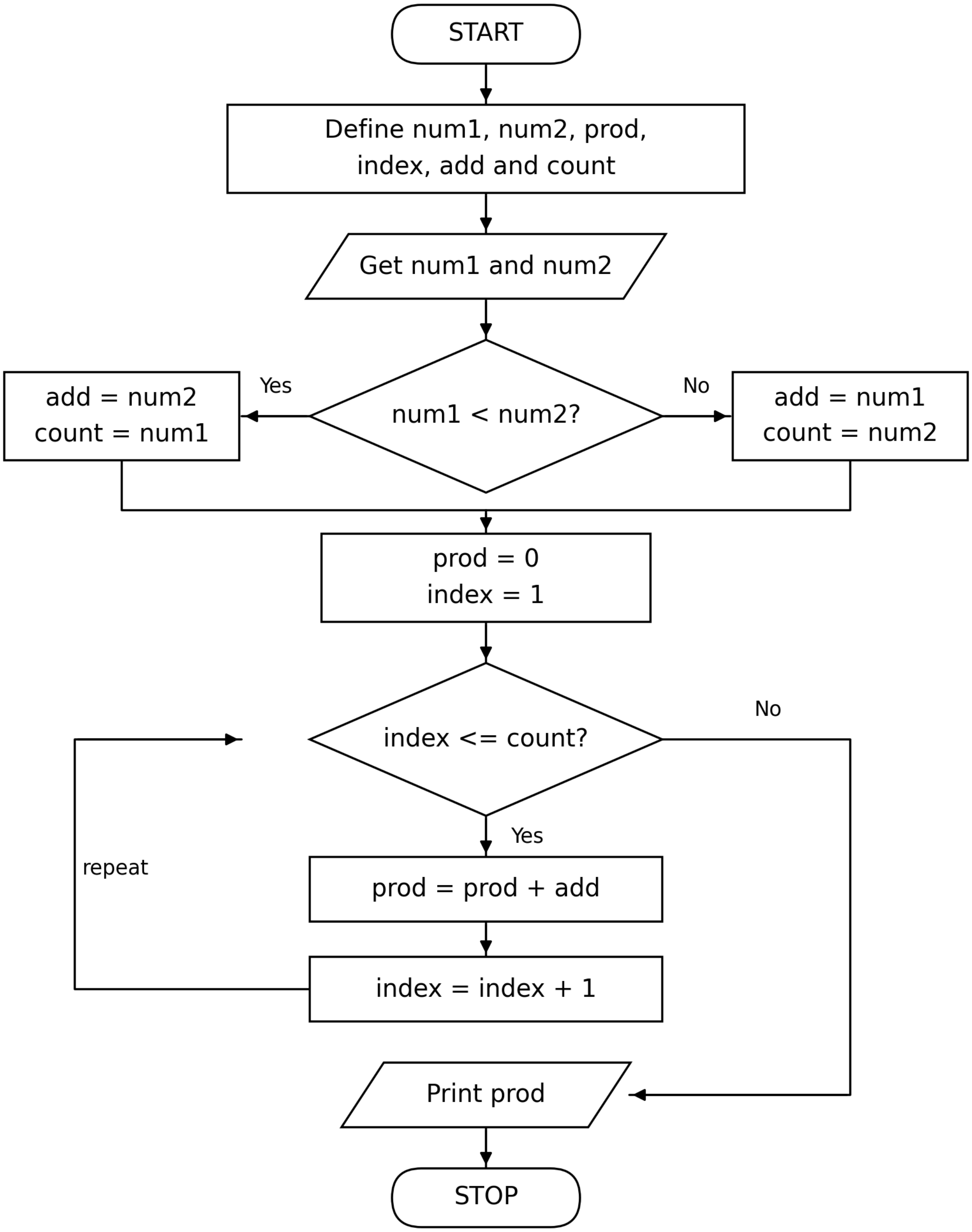
• The original flowchart always adds num1 and repeats based on num2.

• It therefore does not give the minimum possible number of additions.

• The loop condition should use a repetition variable such as count, consistent with the corrected algorithm.

5. Corrected Flowchart

The corrected flowchart first selects the larger number to add and the smaller number as the repetition count. The arrows show the Yes/No paths and the loop back to the decision clearly.



6. Verification

The corrected algorithm and flowchart can be verified by substituting each pair of values given in the assignment. For every pair, the repeated-addition result should match the expected product, and the number of additions should equal the smaller positive input.

The original numerical verification table was not available in the supplied material, so no values have been invented here.

7. Conclusion

The mistakes in the original algorithm and flowchart were identified and corrected. The corrected method performs multiplication through repeated addition without using the multiplication operator. It also reduces the number of additions by using the smaller input as the number of repetitions.